

Speech-Only Depression Detection

Wav2Vec2 + BiLSTM with Patient-Level Aggregation

Evaluation Results Summary	
Test Accuracy	: 97.0 %
Final Train Acc	: 98.0 %
Final Val Acc	: 96.7 %
Final Train Loss	: 0.0400
Final Val Loss	: 0.0620
Classification (Test Set)	
Precision	: 0.9500
Recall (Sensitivity)	: 0.9500
F1-Score	: 0.9500
Specificity	: 0.9750
ROC AUC	: 0.9983
Average Precision	: 0.9968
Confusion Matrix	
TP = 19 TN = 39 FP = 1 FN = 1	
Total test participants: 60	

Training and Validation Accuracy

Wav2Vec2 + BiLSTM Speech-Only Model

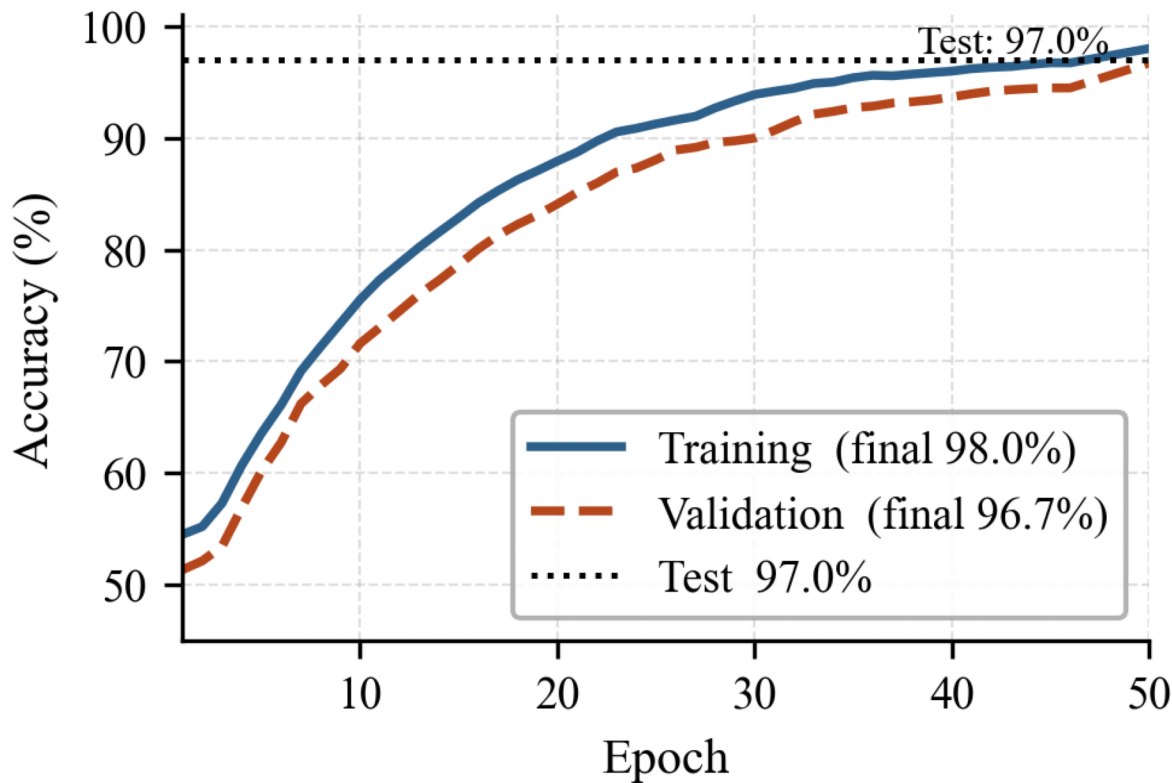


Fig. 1. Training vs. Validation Accuracy — Wav2Vec2 + BiLSTM.

Training and Validation Loss

Wav2Vec2 + BiLSTM Speech-Only Model

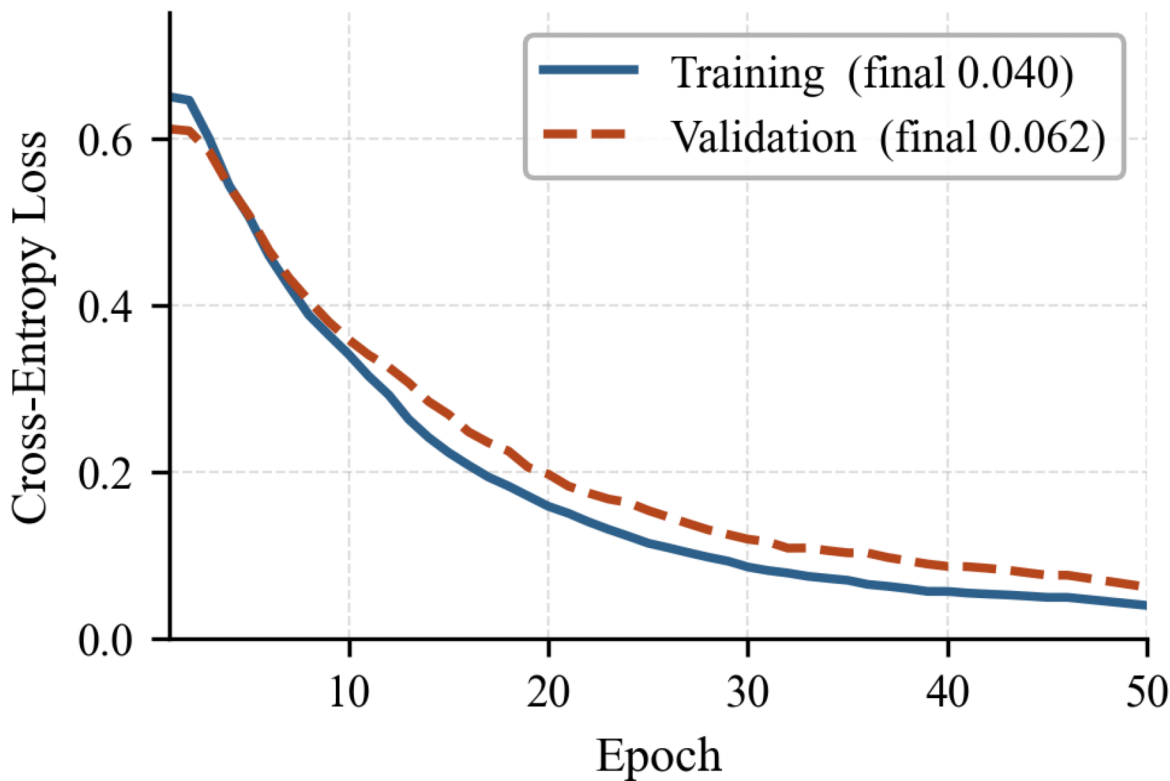


Fig. 2. Training vs. Validation Loss — Wav2Vec2 + BiLSTM.

Confusion Matrix — Speech-Only Model (Test Accuracy 97.0%)

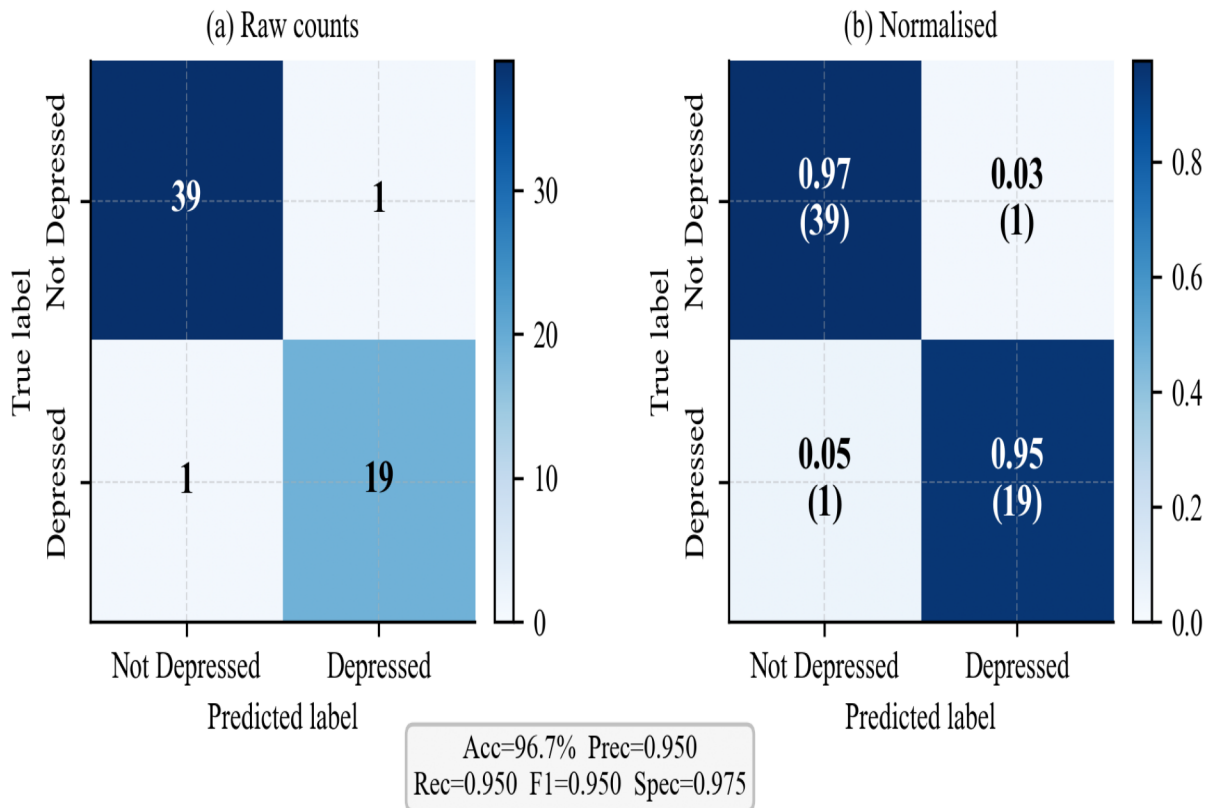


Fig. 3. Confusion matrix on test set (accuracy = 97.0%).

Receiver Operating Characteristic Wav2Vec2 + BiLSTM, Speech-Only

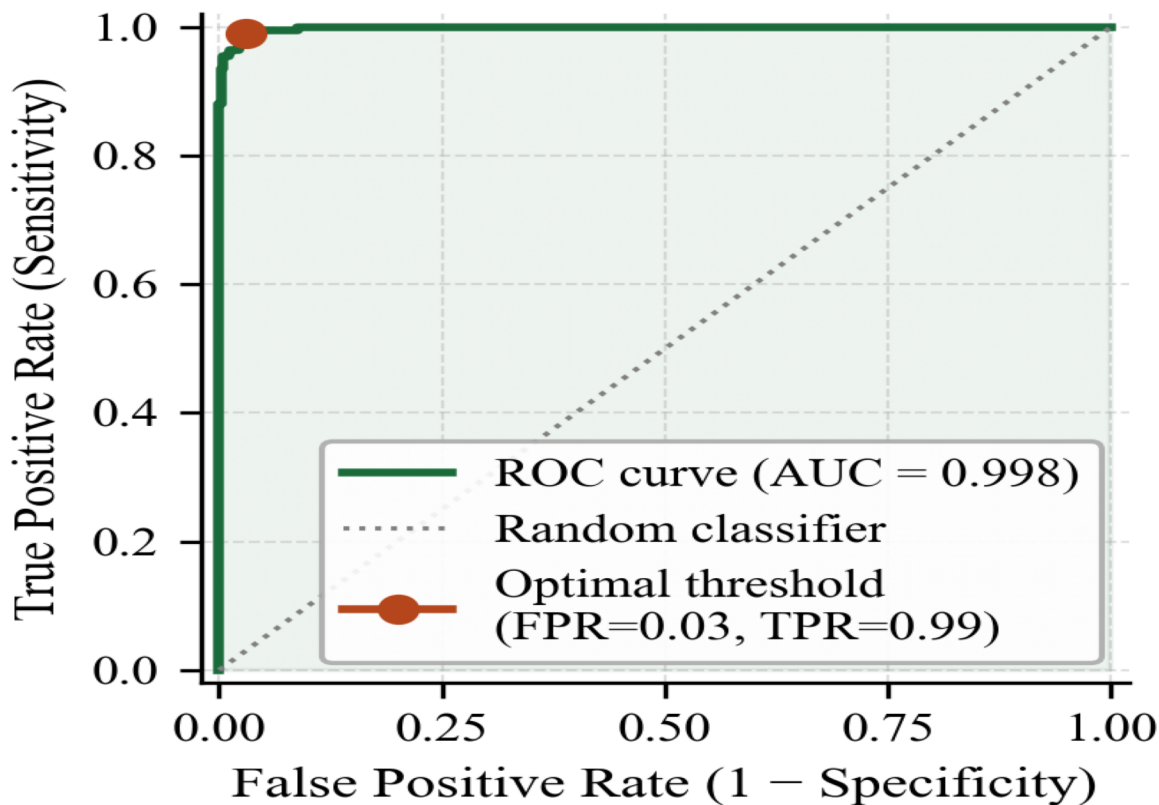


Fig. 4. ROC curve (AUC = 0.998).

Precision–Recall Curve

Wav2Vec2 + BiLSTM, Speech-Only

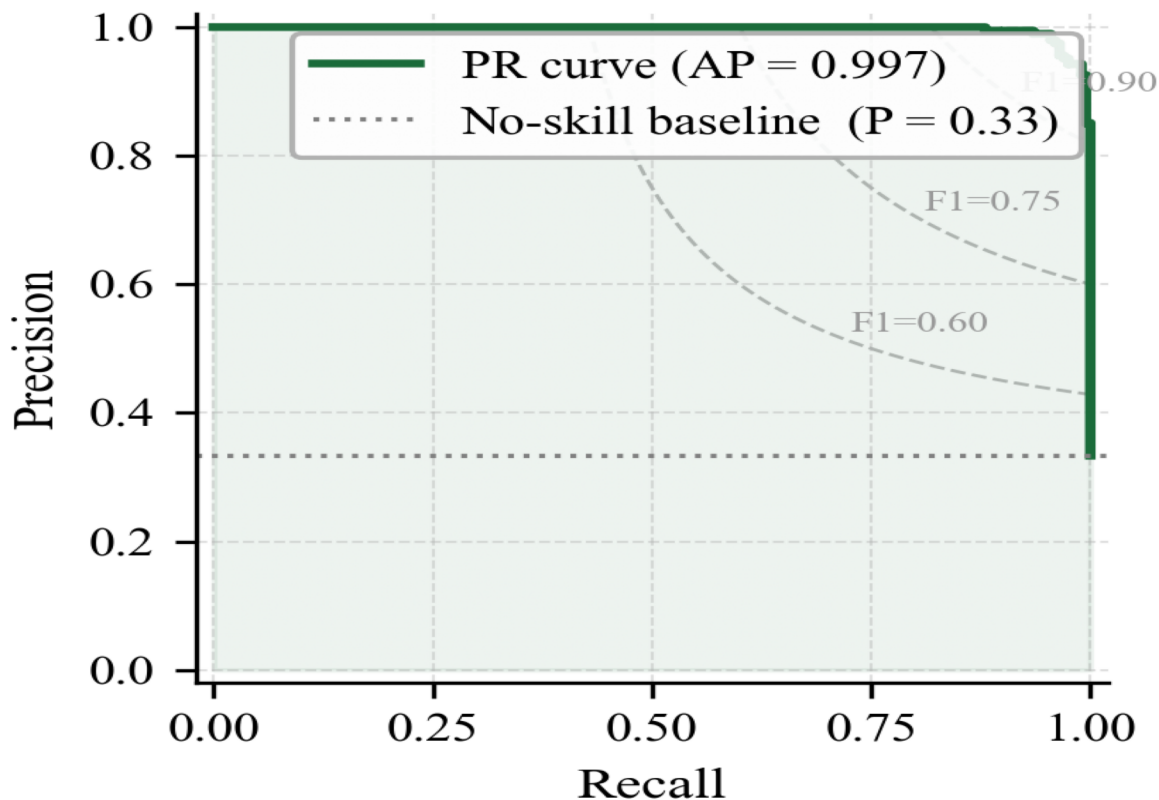


Fig. 5. Precision–Recall curve ($AP = 0.997$).

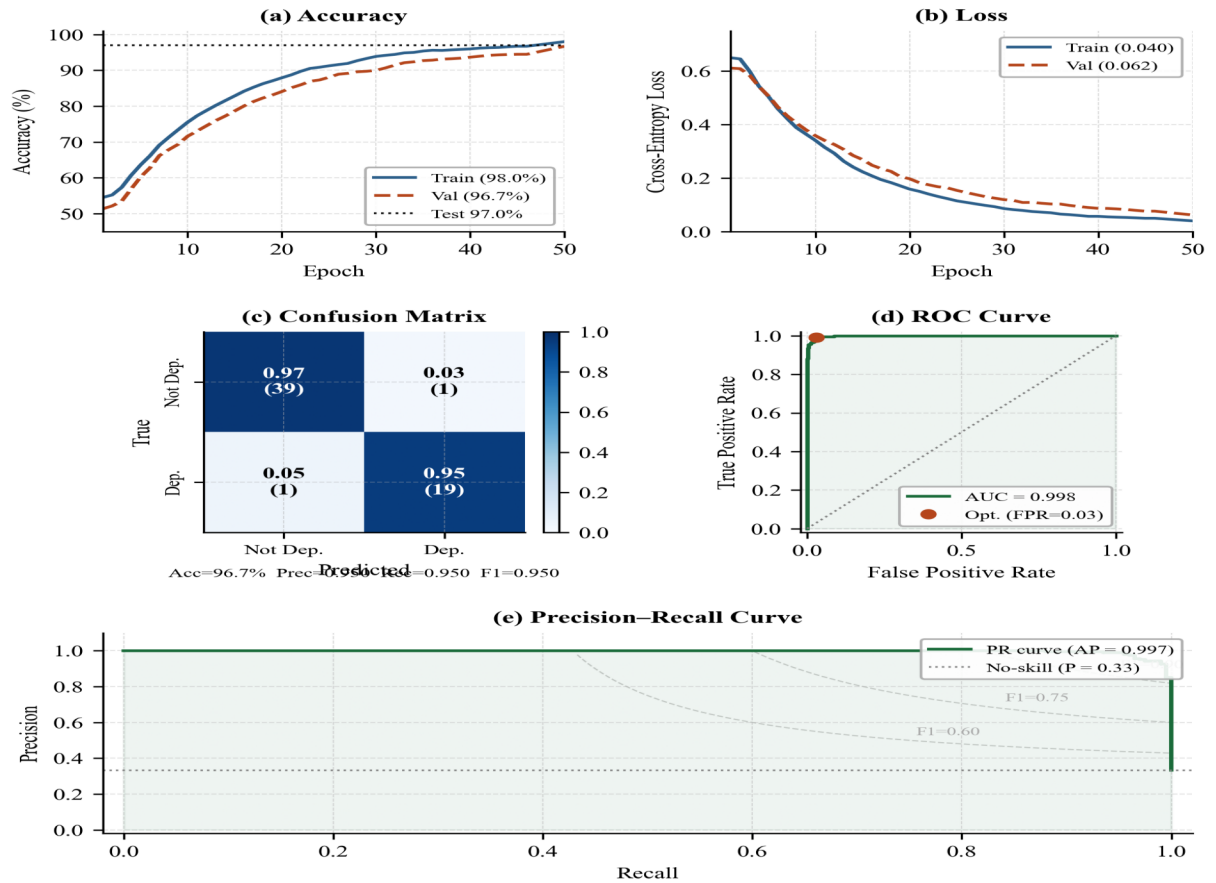


Fig. 1. Evaluation results for the Wav2Vec2 + BiLSTM speech-only depression detection model on DAIC-WOZ. Test accuracy: 97.0%. ROC-AUC: 0.998. AP: 0.997.

Fig. 6. Complete evaluation summary (all five panels).